Dynamics of a Transverse Inerter-Based X-Structure (TIXS) Isolator and Its Application to an HTS Maglev Vehicle Secondary Suspension

1 Scope and conventions

Sections 2 and 3 reconstruct the exact TIXS model in Secs. 2.1–2.2 and Eqs. (1)–(15) of Wang et al. [1]. Sections 4 and 5 derive a new four-degree-of-freedom (4-DOF) vehicle model by replacing the two secondary air-spring branches in the HTS maglev model of Li et al. [2], Fig. 3 and Eqs. (1)–(4), with two TIXS branches. The combined vehicle equations below are a derivation from the two papers, not equations already presented by Li et al.

All vertical displacements below are positive upward. Positive car-body pitch raises the front attachment point. Unless stated otherwise, dynamic coordinates are measured from a loaded static equilibrium, so constant gravity terms cancel from the perturbation equations. The rods and hinges are ideal and massless; car-body roll, lateral motion, bogie pitch, and nonlinear HTS levitation hysteresis are excluded. These assumptions match the scope of the reduced-order models and must be revisited for a full vehicle model.

2 Exact dynamics of one TIXS branch

2.1 Geometric constraint

The TIXS of Wang et al. [1], Figs. 2–3, has four identical rigid rods of length l, a transverse spring of stiffness k, a transverse inerter of inertance b, and a vertical damper of coefficient c. Let y be the upper terminal displacement and z the base displacement. Define their relative vertical displacement by

$$u = y - z. \tag{2.1}$$

The reference rod angle measured from the horizontal is α, and the reference vertical separation is $h_0 = 2l \sin\alpha$. Set

$$\begin{aligned} s(u) &= l\sin\alpha + \frac{u}{2}, \qquad R(u) = \sqrt{l^2 - s(u)^2}, \\ h(u) &= 2s(u) = h_0 + u. \end{aligned} \tag{2.2}$$

Here s and R are the vertical and horizontal projections of one rod. If w is the instantaneous distance between the two lateral hinges and x is its shortening from the reference configuration, the rigid-rod constraint gives

$$\left(l\cos\alpha - \frac{x}{2}\right)^2 + \left(l\sin\alpha + \frac{u}{2}\right)^2 = l^2. \tag{2.3}$$

Therefore,

$$\begin{aligned} w(u) &= 2R(u), \\ x(u) &= 2l\cos\alpha - w(u) \\ &= 2l\cos\alpha - 2R(u). \end{aligned} \tag{2.4}$$

These reproduce Eqs. (1)–(3) of Wang et al. on the physical, non-inverted branch. The allowed range is $0 < h(u) < 2l$, or

$$-2l\sin\alpha < u < 2l(1-\sin\alpha). \tag{2.5}$$

Since $s' = 1/2$ and $R' = -s/(2R)$, direct differentiation yields

$$\begin{aligned} x'(u) &= \frac{s}{R}, \\ x''(u) &= \frac{l^2}{2R^3}. \end{aligned} \tag{2.6}$$

The corresponding velocity and acceleration are

$$\begin{aligned} \dot{x} &= \frac{s}{R}\dot{u}, \\ \ddot{x} &= \frac{s}{R}\ddot{u} + \frac{l^2}{2R^3}\dot{u}^2. \end{aligned} \tag{2.7}$$

Equivalently, the instantaneous rod angle θ satisfies $\sin\theta = s/l$, so $x' = \tan\theta$ and $\dot{w} = -\tan\theta\,\dot{u}$. Replacing $\tan\theta$ by the constant $\tan\alpha$ is a subsequent small-motion approximation, not an exact kinematic identity.

2.2 Kinetic energy, potential energy, and dissipation

Let M_1 be the isolated mass. In the original unpreloaded TIXS model, the spring has zero deformation at $u = 0$. The kinetic energy T, spring potential V, and Rayleigh dissipation function $\mathcal{D}$ are

$$\begin{aligned} T &= \frac{1}{2}M_1\dot{y}^2 + \frac{1}{2}b\dot{x}^2 \\ &= \frac{1}{2}M_1(\dot{u}+\dot{z})^2 + \frac{1}{2}m_X(u)\dot{u}^2, \\ V &= \frac{1}{2}kx(u)^2, \qquad \mathcal{D} = \frac{1}{2}c\dot{u}^2. \end{aligned} \tag{2.8}$$

where the displacement-dependent equivalent vertical inertance is

$$m_X(u) = b[x'(u)]^2 = b\frac{s^2}{l^2 - s^2} = b\tan^2\theta. \tag{2.9}$$

Its derivative is

$$m_X'(u) = \frac{bl^2 s}{(l^2-s^2)^2} = \frac{bl^2 s}{R^4}. \tag{2.10}$$

The inertance b is a two-terminal dynamic parameter. It cannot be added to M_1 as a constant mass without first transforming the transverse motion through the X geometry.

2.3 Lagrange derivation of the exact equation

Use u as the generalized coordinate and $L = T - V$. The required derivatives are

$$\frac{\partial T}{\partial \dot{u}} = M_1(\dot{u} + \dot{z}) + m_X \dot{u}, \tag{2.11}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{u}} = M_1(\ddot{u} + \ddot{z}) + m_X \ddot{u} + m'_X \dot{u}^2, \tag{2.12}$$

$$\frac{\partial T}{\partial u} = \frac{1}{2} m'_X \dot{u}^2, \tag{2.13}$$

$$\frac{\partial V}{\partial u} = kxx', \tag{2.14}$$

$$\frac{\partial \mathcal{D}}{\partial \dot{u}} = c\dot{u}. \tag{2.15}$$

Inserting these into the Lagrange equation $\mathrm{d}(\partial L/\partial \dot{u})/\mathrm{d}t - \partial L/\partial u + \partial \mathcal{D}/\partial \dot{u} = 0$ gives

$$\boxed{[M_1 + m_X(u)]\ddot{u} + \frac{1}{2} m'_X(u)\dot{u}^2 + c\dot{u} + kx(u)x'(u) = -M_1\ddot{z}.} \tag{2.16}$$

Substituting Eqs. (2.4), (2.6), (2.9), and (2.10) gives the explicit form corresponding to Eq. (9) of Wang et al.:

$$\boxed{\begin{aligned}&\left(M_1 + b\frac{s^2}{l^2 - s^2}\right)\ddot{u} + \frac{bl^2 s}{2(l^2 - s^2)^2}\dot{u}^2 + c\dot{u} \\ &\quad + k\left(2l\cos\alpha - 2\sqrt{l^2 - s^2}\right)\frac{s}{\sqrt{l^2 - s^2}} = -M_1\ddot{z}.\end{aligned}} \tag{2.17}$$

The second term is a geometry-induced velocity-squared term. It may change sign with s and must be retained in the exact nonlinear equation. As an independent check, the transverse inerter force is $b\ddot{x}$. Virtual-work transformation gives the vertical generalized force $b\ddot{x}\,x' = m_X\ddot{u} + bx'x''\dot{u}^2$, with $bx'x'' = m'_X/2$, exactly as in Eq. (2.16).

2.4 Connection to the paper's dimensionless equation

Define

$$\begin{aligned}\omega_n &= \sqrt{k/M_1}, \quad U = \frac{u}{2l}, \qquad Z = \frac{z_m}{2l}, \\ \tau &= \omega_n t, \qquad \Omega = \frac{\omega}{\omega_n}, \\ \delta &= \frac{b}{M_1}, \qquad \zeta = \frac{c}{2M_1\omega_n}.\end{aligned} \tag{2.18}$$

For harmonic base motion $z = z_m \cos\omega t$, let primes denote differentiation with respect to τ and put $a = U + \sin\alpha$. Dividing Eq. (2.17) by $2lk$ yields

$$\boxed{\begin{aligned}&U'' + \delta\frac{a^2}{1 - a^2}U'' + \delta\frac{a}{(1 - a^2)^2}(U')^2 + 2\zeta U' \\ &\qquad + \frac{(\cos\alpha - \sqrt{1 - a^2})a}{\sqrt{1 - a^2}} = Z\Omega^2\cos(\Omega\tau).\end{aligned}} \tag{2.19}$$

This is the exact-geometry equation corresponding to Eqs. (10)–(15) of Wang et al. Their later third-order Taylor approximations are not used in the vehicle derivation.

3 The four-DOF HTS maglev vehicle before replacement

Li et al. [2] model the car body by its vertical translation and pitch and model each bogie–Dewar assembly by one vertical translation. Let the car-body mass and pitch inertia be m_2, I_2, the front and rear bogie masses be m_{1a}, m_{1b}, and the distances from the car-body center to the front and rear secondary-suspension attachment points be L_a, L_b. Primary HTS levitation is linearized with stiffnesses K_{1i} and damping coefficients C_{1i}; guideway inputs are ξ_a, ξ_b.

For a consistent upward coordinate system, denote the car-body, front-bogie, and rear-bogie vertical motions by z_c, z_a, z_b, and pitch by φ. Write F_{2i} for the downward restoring-force component applied to the car body by secondary branch i, and p_i for the upward actuator force on the car body. Newton's equations, with the same force topology as Eqs. (1)–(4) of Li et al., are

$$m_2\ddot{z}_c + F_{2a} + F_{2b} = p_a + p_b, \tag{3.1}$$
$$I_2\ddot{\varphi} + L_aF_{2a} - L_bF_{2b} = L_ap_a - L_bp_b, \tag{3.2}$$
$$m_{1a}\ddot{z}_a + P_a - F_{2a} = -p_a, \tag{3.3}$$
$$m_{1b}\ddot{z}_b + P_b - F_{2b} = -p_b. \tag{3.4}$$

With linear air-spring secondary suspension, these force components are

$$\begin{aligned} F_{2i}^{\mathrm{lin}} &= K_{2i}r_i + C_{2i}\dot{r}_i, \\ P_i &= K_{1i}(z_i - \xi_i) + C_{1i}(\dot{z}_i - \dot{\xi}_i). \end{aligned} \tag{3.5}$$

The F_{k2i} and F_{c2i} of Li et al. correspond to the elastic and viscous parts of F_{2i}^{lin}, and their active force f_{ci} corresponds to p_i. Their Fig. 3 uses differing arrow directions for some displacements; the definitions above fix all signs explicitly. In the passive case $p_a = p_b = 0$. Retaining the actuators leaves their equal-and-opposite forces in Eqs. (3.1)–(3.4).

4 TIXS replacement of the secondary suspension

4.1 Attachment kinematics

Define the generalized coordinate and two constant incidence vectors by

$$\begin{aligned} \boldsymbol{q} &= (z_c, \varphi, z_a, z_b)^{\mathsf{T}}, \\ \boldsymbol{B}_a &= (1, L_a, -1, 0)^{\mathsf{T}}, \\ \boldsymbol{B}_b &= (1, -L_b, 0, -1)^{\mathsf{T}}. \end{aligned} \tag{4.1}$$

For small rigid-body pitch, the front and rear car-body attachment displacements are $z_c + L_a\varphi$ and $z_c - L_b\varphi$. Thus

$$r_a = z_c + L_a\varphi - z_a = \boldsymbol{B}_a^{\mathsf{T}}\boldsymbol{q}, \tag{4.2}$$
$$r_b = z_c - L_b\varphi - z_b = \boldsymbol{B}_b^{\mathsf{T}}\boldsymbol{q}. \tag{4.3}$$

For either branch $i \in \{a, b\}$,

$$\dot{r}_i = \boldsymbol{B}_i^{\mathsf{T}} \dot{\boldsymbol{q}}, \qquad \ddot{r}_i = \boldsymbol{B}_i^{\mathsf{T}} \ddot{\boldsymbol{q}}. \tag{4.4}$$

The small-angle approximation applies only to attachment-point kinematics. The X-structure geometry within each TIXS remains exact over its admissible stroke.

4.2 Static load and necessary spring preload

An important difference arises when an air spring is fully replaced. The unpreloaded spring of the original TIXS has zero vertical elastic force at $r_i = 0$. Two such branches alone cannot support a nonzero car-body weight at that reference configuration. One must either retain a separate static-load path or preload the transverse springs. We use preload and define, for $i \in \{a, b\}$,

$$\begin{aligned} s_i &= l_i \sin\alpha_i + \frac{r_i}{2}, & R_i &= \sqrt{l_i^2 - s_i^2}, \\ w_i(r_i) &= 2R_i, & w_{0i} &= 2l_i \cos\alpha_i, \\ V_i(r_i) &= \frac{1}{2} k_i [w_i(r_i) - w_{ni}]^2. \end{aligned} \tag{4.5}$$

Here w_{ni} is the natural transverse spring length. The generalized vertical spring force at the static operating point is

$$F_{0i} = V_i'(0) = -k_i (w_{0i} - w_{ni}) \tan\alpha_i. \tag{4.6}$$

If these are the only two branches supporting the car body and there is no externally applied static pitch moment, the car-body static force and moment balances give

$$\begin{aligned} F_{0a} + F_{0b} &= -m_2 g, \\ L_a F_{0a} - L_b F_{0b} &= 0, \\ F_{0a} &= -\frac{m_2 g L_b}{L_a + L_b}, \\ F_{0b} &= -\frac{m_2 g L_a}{L_a + L_b}. \end{aligned} \tag{4.7}$$

For a specified design load, $w_{ni} = w_{0i} + F_{0i}/(k_i \tan\alpha_i)$ sets the required preload. Its physical feasibility requires $w_{ni} > 0$, adequate spring travel, and strength checks. If the original unpreloaded TIXS is used instead, a separate static-load path is necessary. The primary levitation supports the bogie weights and the static forces passed down from the car body; only perturbations about that complete equilibrium appear below.

4.3 Two-terminal inertial, elastic, and damping forces

The transverse inerter energy of branch i is

$$\begin{aligned} T_{Xi} &= \frac{1}{2} b_i \dot{w}_i^2 = \frac{1}{2} \mu_i(r_i) \dot{r}_i^2, \\ \mu_i(r_i) &= b_i \frac{s_i^2}{R_i^2}, \\ \mu_i'(r_i) &= \frac{b_i l_i^2 s_i}{R_i^4}. \end{aligned} \tag{4.8}$$

The damper contributes $\mathcal{D}_i = c_i\dot{r}_i^2/2$. Lagrange's equation in the relative branch coordinate gives its dynamic force after subtracting the static preload:

$$\boxed{\Delta F_i = \mu_i(r_i)\ddot{r}_i + \frac{1}{2}\mu_i'(r_i)\dot{r}_i^2 + c_i\dot{r}_i + V_i'(r_i) - F_{0i}.} \tag{4.9}$$

The spring term is

$$\begin{aligned} w_i' &= -\frac{s_i}{R_i}, \\ V_i'(r_i) &= k_i(w_i - w_{ni})w_i' \\ &= -k_i(2R_i - w_{ni})\frac{s_i}{R_i}. \end{aligned} \tag{4.10}$$

Consequently, the exact force can also be written without implicit derivatives:

$$\boxed{\begin{aligned} \Delta F_i = {} & b_i\frac{s_i^2}{l_i^2 - s_i^2}\ddot{r}_i + \frac{b_i l_i^2 s_i}{2(l_i^2 - s_i^2)^2}\dot{r}_i^2 + c_i\dot{r}_i \\ & - k_i\left(2\sqrt{l_i^2 - s_i^2} - w_{ni}\right)\frac{s_i}{\sqrt{l_i^2 - s_i^2}} - F_{0i}. \end{aligned}} \tag{4.11}$$

When $w_{ni} = w_{0i}$, $F_{0i} = 0$ and the spring term becomes $k_i(2l_i\cos\alpha_i - 2R_i)s_i/R_i$, agreeing with the isolated TIXS equation. The inerter acts between two moving terminals, so its energy belongs to the coupled vehicle rather than to either terminal mass alone.

4.4 Vehicle Lagrangian and four scalar equations

Let $\boldsymbol{e}_a = (0,0,1,0)^{\mathsf{T}}$ and $\boldsymbol{e}_b = (0,0,0,1)^{\mathsf{T}}$. The total kinetic energy, perturbation potential, and Rayleigh function are

$$\begin{aligned} T = {} & \frac{1}{2}m_2\dot{z}_c^2 + \frac{1}{2}I_2\dot{\varphi}^2 + \frac{1}{2}m_{1a}\dot{z}_a^2 + \frac{1}{2}m_{1b}\dot{z}_b^2 \\ & + \sum_{i=a,b}\frac{1}{2}\mu_i(r_i)\dot{r}_i^2, \end{aligned} \tag{4.12}$$

$$\begin{aligned} \widetilde{V} = {} & \sum_{i=a,b}\Big[V_i(r_i) - V_i(0) - F_{0i}r_i \\ & + \frac{1}{2}K_{1i}(z_i - \xi_i)^2\Big], \end{aligned} \tag{4.13}$$

$$\mathcal{D} = \sum_{i=a,b}\left[\frac{1}{2}c_i\dot{r}_i^2 + \frac{1}{2}C_{1i}(\dot{z}_i - \dot{\xi}_i)^2\right]. \tag{4.14}$$

Subtracting $F_{0i}r_i$ from the branch potential is the result of expanding the full gravitational and elastic potential about the loaded equilibrium. If absolute coordinates are used instead, the car-body and bogie gravity potentials and primary-levitation static forces must be included explicitly before the equilibrium terms are canceled.

Virtual work uses $\delta r_i = \boldsymbol{B}_i^{\mathsf{T}}\delta\boldsymbol{q}$. Thus a branch force contributes $\boldsymbol{B}_i\Delta F_i$ to the generalized restoring force. Lagrange's equations become

$$\boxed{\begin{aligned} & \boldsymbol{M}_0\ddot{\boldsymbol{q}} + \sum_{i=a,b}\boldsymbol{B}_i\Delta F_i + \boldsymbol{e}_a P_a + \boldsymbol{e}_b P_b \\ & \qquad = \boldsymbol{B}_a p_a + \boldsymbol{B}_b p_b. \end{aligned}} \tag{4.15}$$

Here $\boldsymbol{M}_0 = \mathrm{diag}(m_2, I_2, m_{1a}, m_{1b})$. Expanding each degree of freedom gives

$$m_2\ddot{z}_c + \Delta F_a + \Delta F_b = p_a + p_b, \tag{4.16}$$
$$I_2\ddot{\varphi} + L_a\Delta F_a - L_b\Delta F_b = L_a p_a - L_b p_b, \tag{4.17}$$
$$m_{1a}\ddot{z}_a + P_a - \Delta F_a = -p_a, \tag{4.18}$$
$$m_{1b}\ddot{z}_b + P_b - \Delta F_b = -p_b. \tag{4.19}$$

These have the same attachment topology as Li et al.'s four original equations. The linear secondary force has been replaced by the exact nonlinear TIXS force ΔF_i. Because that force contains $\ddot{r}_i$, Eqs. (4.16)–(4.19) are coupled implicitly in the accelerations. The next form is directly suitable for numerical integration.

4.5 State-dependent vehicle mass matrix

Separate the acceleration-dependent part of Eq. (4.9) by defining

$$\begin{aligned} H_i(r_i, \dot{r}_i) &= \frac{1}{2}\mu_i'(r_i)\dot{r}_i^2 + c_i\dot{r}_i + V_i'(r_i) - F_{0i}, \\ P_i &= K_{1i}(z_i - \xi_i) + C_{1i}(\dot{z}_i - \dot{\xi}_i). \end{aligned} \tag{4.20}$$

Since $\ddot{r}_i = \boldsymbol{B}_i^{\mathsf{T}}\ddot{\boldsymbol{q}}$, moving $\mu_i\ddot{r}_i$ to the mass matrix yields

$$\boxed{\begin{aligned} \boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} = \boldsymbol{B}_a p_a + \boldsymbol{B}_b p_b \\ - \boldsymbol{B}_a H_a - \boldsymbol{B}_b H_b \\ - \boldsymbol{e}_a P_a - \boldsymbol{e}_b P_b. \end{aligned}} \tag{4.21}$$

$$\boldsymbol{M}(\boldsymbol{q}) = \boldsymbol{M}_0 + \sum_{i=a,b} \mu_i(r_i)\boldsymbol{B}_i\boldsymbol{B}_i^{\mathsf{T}}. \tag{4.22}$$

Writing $\mu_a = \mu_a(r_a)$ and $\mu_b = \mu_b(r_b)$, all entries are

$$\boldsymbol{M} = \begin{bmatrix} m_2 + \mu_a + \mu_b & L_a\mu_a - L_b\mu_b & -\mu_a & -\mu_b \\ L_a\mu_a - L_b\mu_b & I_2 + L_a^2\mu_a + L_b^2\mu_b & -L_a\mu_a & L_b\mu_b \\ -\mu_a & -L_a\mu_a & m_{1a} + \mu_a & 0 \\ -\mu_b & L_b\mu_b & 0 & m_{1b} + \mu_b \end{bmatrix}. \tag{4.23}$$

For positive masses and $b_i \geq 0$, the matrix $\boldsymbol{M}$ is symmetric positive definite when $R_i > 0$. During integration, compute $r_i, s_i, R_i, \mu_i, H_i, P_i$ from the current state and solve Eq. (4.21) for $\ddot{\boldsymbol{q}}$. Neither the state dependence of μ_i nor the term $\mu_i'\dot{r}_i^2/2$ may be omitted in the exact model.

4.6 Small-amplitude limit and consistency checks

About $r_i = \dot{r}_i = 0$, the velocity-squared term is second order. The equilibrium equivalent inertance is

$$\mu_{i0} = b_i \tan^2\alpha_i. \tag{4.24}$$

The tangent stiffness follows from $V_i'' = k_i[(w_i')^2 + (w_i - w_{ni})w_i'']$, with $w_i'(0) = -\tan\alpha_i$ and $w_i''(0) = -1/(2l_i\cos^3\alpha_i)$:

$$K_{Ti} = V_i''(0) = k_i\left[\tan^2\alpha_i - \frac{w_{0i} - w_{ni}}{2l_i\cos^3\alpha_i}\right]. \tag{4.25}$$

Hence $\Delta F_i \simeq \mu_{i0}\ddot{r}_i + c_i\dot{r}_i + K_{Ti}r_i$ to first order. Without preload, $w_{ni} = w_{0i}$ and $K_{Ti} = k_i\tan^2\alpha_i$. If additionally $b_i = 0$, the vehicle equations reduce to the conventional linear secondary spring-damper form with $K_{2i} = K_{Ti}$ and $C_{2i} = c_i$. When $b_i > 0$, the linearized vehicle still contains the coupled added-mass term $\sum_i \mu_{i0}\boldsymbol{B}_i\boldsymbol{B}_i^{\mathsf{T}}$, so matching secondary stiffness and damping alone does not reproduce the original air-spring vehicle. Preload can reduce K_{Ti}; static stability and full-stroke behavior require separate checks.

5 Limits of the derived model

"Exact" refers here to the rigid-rod geometry and two-terminal inertance transformation without Taylor truncation. It does not account for rod mass, joint friction, end stops, lateral coupling, or a nonlinear hysteretic HTS levitation law. Each branch must remain within $0 < 2s_i < 2l_i$; as $R_i \to 0$, its geometric derivatives and equivalent inertance diverge. The primary suspension is still the linearized K_{1i}, C_{1i} model of Li et al. Set $p_a = p_b = 0$ for passive TIXS operation. A PID or fuzzy-PID control law may be added through p_i, but it is not implied by the mechanical derivation.

References

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